TRAINITY PROJECT 5

IMDB Movie Analysis

Project Description: This project analyses IMDB movie data to identify key factors influencing high ratings. Through data cleaning, exploration, and the Five 'Whys' approach, we uncover trends in budget, genre, and production quality. Insights help producers, directors, and investors make informed decisions for future projects.

Data Analytics Tasks:

A. **Movie Genre Analysis:** Analyze the distribution of movie genres and their impact on the IMDB score.

- **Task:** Determine the most common genres of movies in the dataset. Then, for each genre, calculate descriptive statistics (mean, median, mode, range, variance, standard deviation) of the IMDB scores.

B. Movie Duration Analysis: Analyze the distribution of movie durations and its impact on the IMDB score.

- Task: Analyze the distribution of movie durations and identify the relationship between movie duration and IMDB score.

C. Language Analysis: Situation: Examine the distribution of movies based on their language.

- **Task:** Determine the most common languages used in movies and analyze their impact on the IMDB score using descriptive statistics.

D. Director Analysis: Influence of directors on movie ratings.

- Task: Identify the top directors based on their average IMDB score and analyze their contribution to the success of movies using percentile calculations.

E. Budget Analysis: Explore the relationship between movie budgets and their financial success.

- Task: Analyze the correlation between movie budgets and gross earnings, and identify the movies with the highest profit margin.

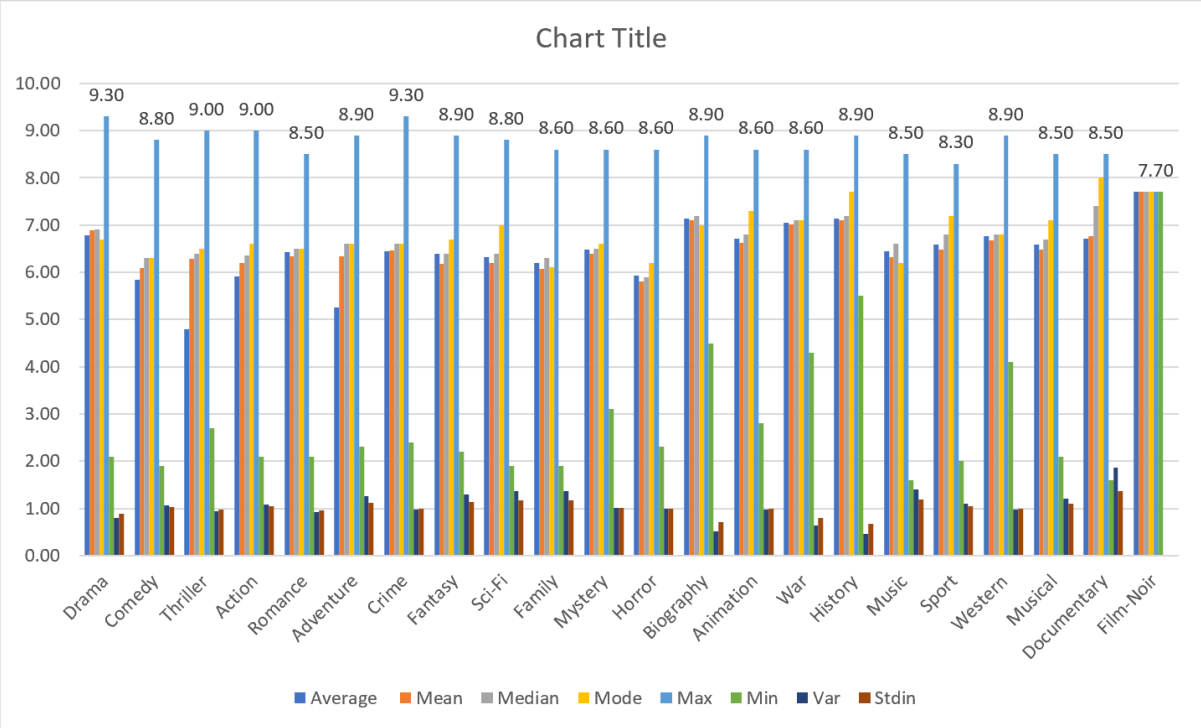
Tools Used: Excel, Trainity Platform

Data Cleaning:

1. The movie title column had 126 duplicate values which was deleted from the dataset.
2. Movie titles had 'Â' in their titles which were removed.
3. There were a lot of blanks present in the data which were filled by searching for information online and if no information was available the row was deleted (a lot of the rows had to be deleted as finding the information was difficult).

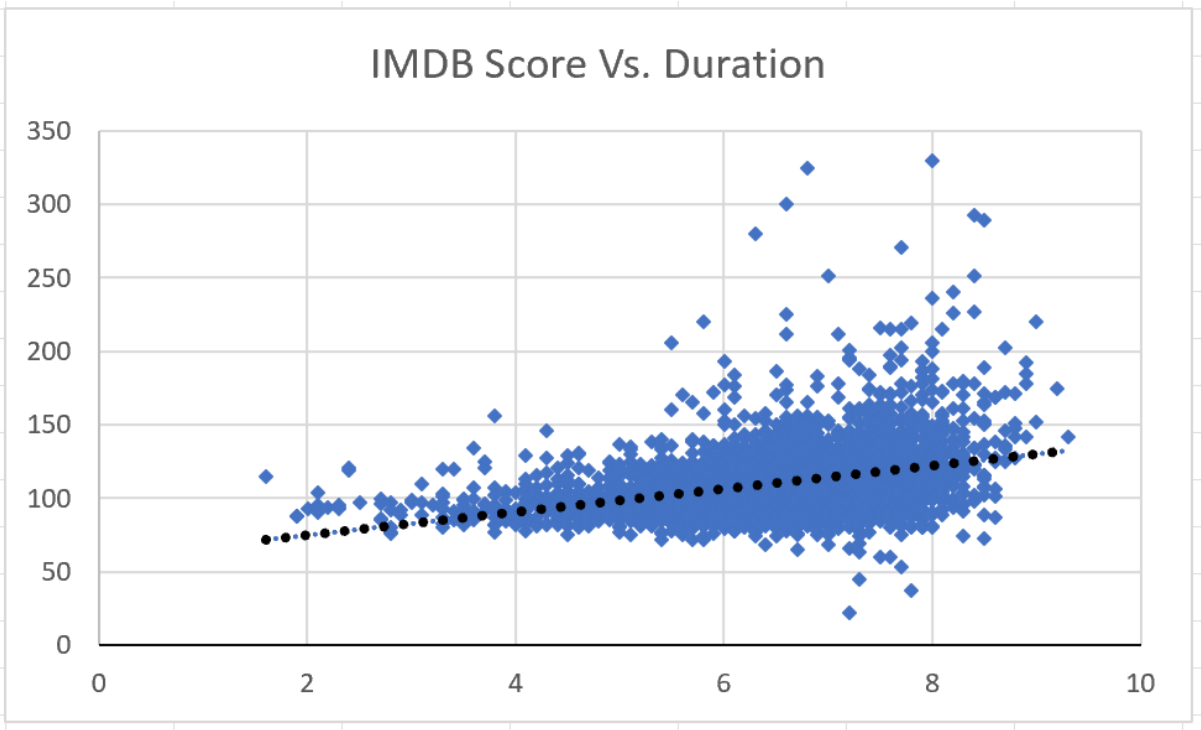
Task 1: Determine the most common genres of movies in the dataset. Then, for each genre, calculate descriptive statistics (mean, median, mode, range, variance, standard deviation) of the IMDB scores.

Genre	Average	Median	Mode	Max	Min	Var	Stdin
Drama	6.79	6.90	6.70	9.30	2.10	0.80	0.89
Comedy	5.84	6.30	6.30	8.80	1.90	1.07	1.03
Thriller	4.80	6.40	6.50	9.00	2.70	0.94	0.97
Action	5.92	6.35	6.60	9.00	2.10	1.09	1.04
Romance	6.43	6.50	6.50	8.50	2.10	0.92	0.96
Adventure	5.25	6.60	6.60	8.90	2.30	1.26	1.12
Crime	6.45	6.60	6.60	9.30	2.40	0.97	0.99
Fantasy	6.40	6.40	6.70	8.90	2.20	1.29	1.14
Sci-Fi	6.33	6.40	7.00	8.80	1.90	1.37	1.17
Family	6.20	6.30	6.10	8.60	1.90	1.37	1.17
Mystery	6.47	6.50	6.60	8.60	3.10	1.01	1.00
Horror	5.93	5.90	6.20	8.60	2.30	0.99	0.99
Biography	7.14	7.20	7.00	8.90	4.50	0.51	0.71
Animation	6.71	6.80	7.30	8.60	2.80	0.98	0.99
War	7.05	7.10	7.10	8.60	4.30	0.64	0.80
History	7.13	7.20	7.70	8.90	5.50	0.46	0.68
Music	6.45	6.60	6.20	8.50	1.60	1.41	1.19
Sport	6.59	6.80	7.20	8.30	2.00	1.09	1.05
Western	6.77	6.80	6.80	8.90	4.10	0.98	0.99
Musical	6.59	6.70	7.10	8.50	2.10	1.21	1.10
Documentary	6.71	7.40	8.00	8.50	1.60	1.85	1.36
Film-Noir	7.70	7.70	7.70	7.70	7.70	0.00	0.00



- 'Film_noir' has the highest average but there is only one recorded movie for this category therefore the highest average IDDB score was for 'Biography' followed closely by 'History'.

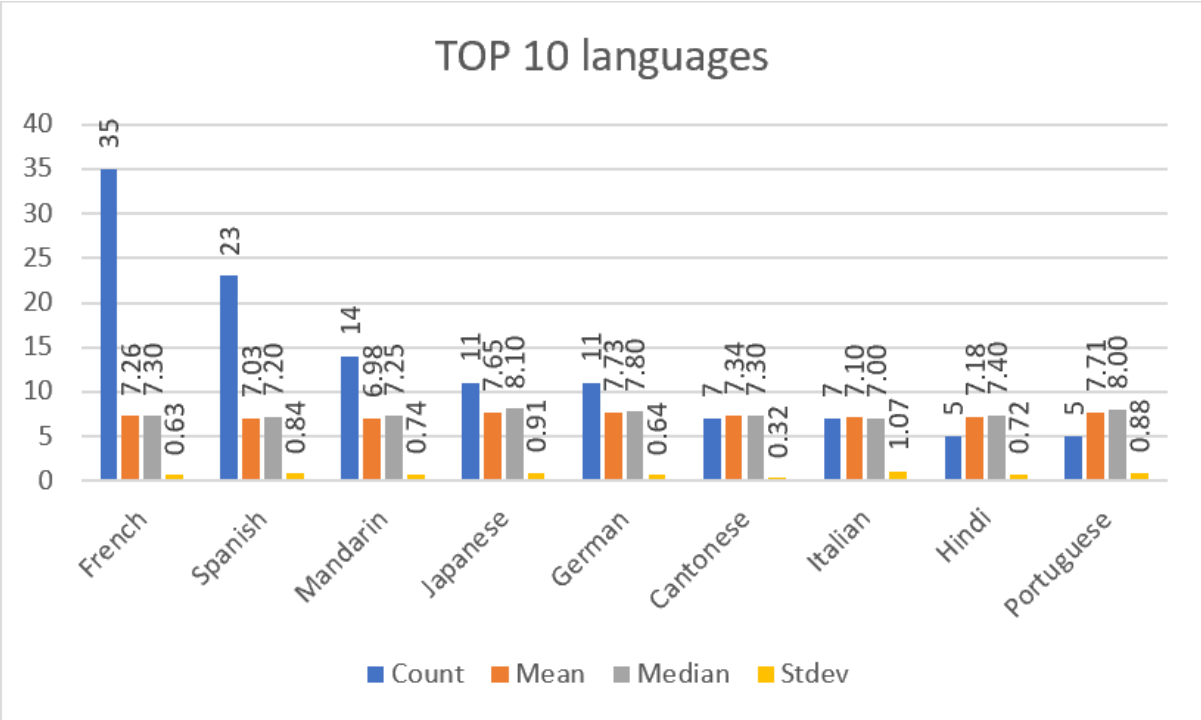
Task 2: Analyze the distribution of movie durations and identify the relationship between movie duration and IMDB score.



- By plotting a scatter plot of IMDB score Vs. Duration of the movies, we can see that there is a slightly increasing trendline associated here indicating that movies with a higher rating are likely to have a longer duration that movies with a lower rating not by a large margin though as the trendline is only ever so slightly increasing.

Task 3: Determine the most common languages used in movies and analyze their impact on the IMDB score using descriptive statistics.

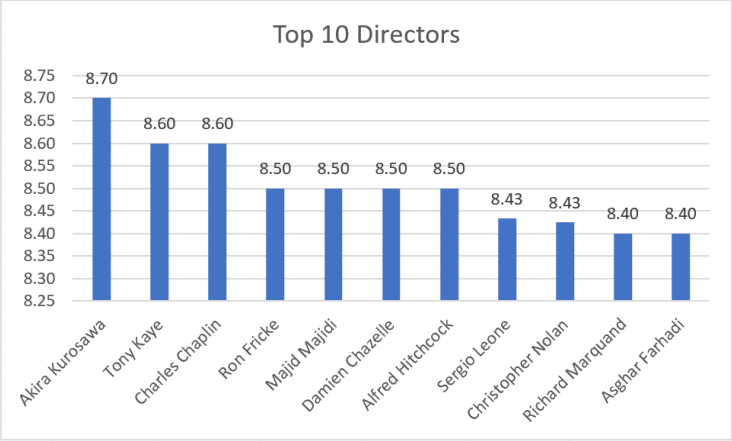
Language	Count	Mean	Median	Stdev
English	3534	6.95	6.50	1.05
French	35	7.26	7.30	0.63
Spanish	23	7.03	7.20	0.84
Mandarin	14	6.98	7.25	0.74
Japanese	11	7.65	8.10	0.91
German	11	7.73	7.80	0.64
Cantonese	7	7.34	7.30	0.32
Italian	7	7.10	7.00	1.07
Hindi	5	7.18	7.40	0.72
Portuguese	5	7.71	8.00	0.88



- The top 10 languages were plotted with English being highest with a big margin followed by French and Spanish with significantly lower values.
- English was not added to the chart due to its really high count, impacting the analysis of the chart.

Task 5: Identify the top directors based on their average IMDB score and analyze their contribution to the success of movies using percentile calculations.

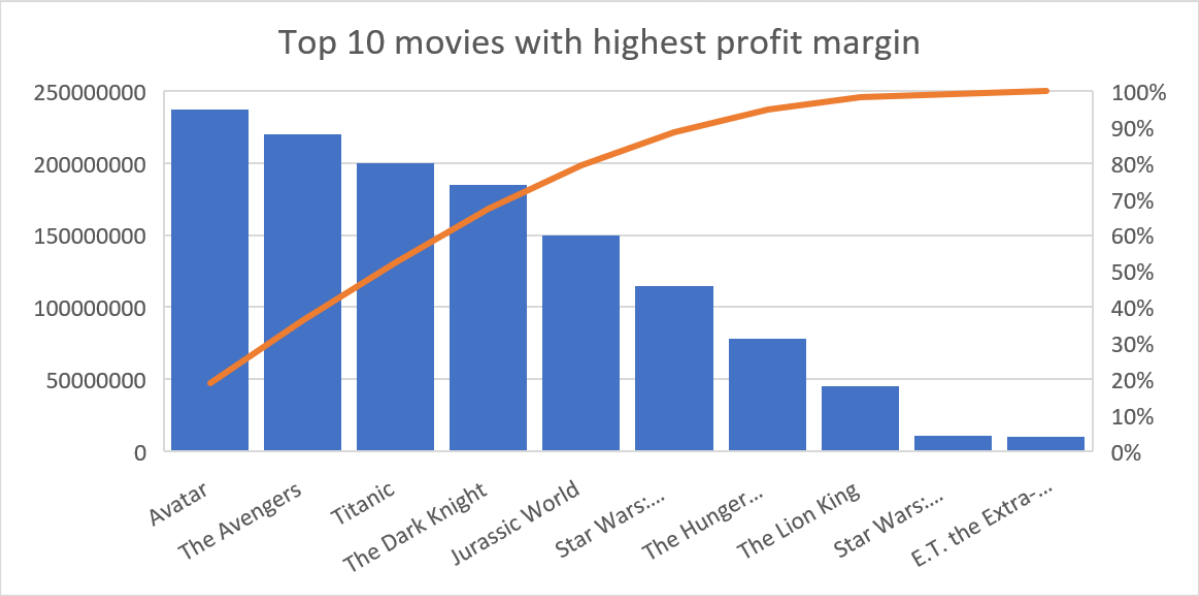
Director	Avg IMDB	Percentile
Akira Kurosawa	8.70	99.90%
Tony Kaye	8.60	99.80%
Charles Chaplin	8.60	99.80%
Ron Fricke	8.50	99.50%
Majid Majidi	8.50	99.50%
Damien Chazelle	8.50	99.50%
Alfred Hitchcock	8.50	99.50%
Sergio Leone	8.43	99.50%
Christopher Nolan	8.43	99.40%
Richard Marquand	8.40	99.30%
Asghar Farhadi	8.40	99.30%



- The top Director was Akira Kurosawa with the highest average rating on his movies of 8.70, followed by a tie between Tony Kaye and Charles Chaplin.
- The average rating of the top 10 directors were very close to each other with very high percentiles, each having a percentile greater than 99%

Task 5: Analyze the correlation between movie budgets and gross earnings, and identify the movies with the highest profit margin.

S.no	movie_title	budget	gross	Profit Margin
1	Avatar	237000000	760505847	449935665
2	Jurassic World	150000000	652177271	424449459
3	Titanic	200000000	658672302	403279547
4	Star Wars: Episode IV - A New Hope	11000000	460935665	377783777
5	E.T. the Extra-Terrestrial	10500000	434949459	359544677
6	The Avengers	220000000	623279547	348316061
7	The Lion King	45000000	422783777	329999255
8	Star Wars: Episode I - The Phantom Menace	115000000	474544677	305024263
9	The Dark Knight	185000000	533316061	294645577
10	The Hunger Games	78000000	407999255	293784000



- Avatar was the highest grossing film and had the highest profit margin as well followed by Jurassic World.
- Even though The Avengers movie was one of the higher grossing movies, due to its high budget it ended up at 6th. Same result was noticed for The Dark Knight which ended up 9th.

Result: This project was quite challenging in terms of figuring out how to map the IMDB scores to movie titles, directors and languages. I learned the concept of using the FILTER formula in excel along with the SEARCH formula for mapping the IMDB scores to respective metrics. I also learned a little about the art of visualization by exploring different charts and understanding which would lead to the best story-telling for the analysed data.

Excel Link:
https://drive.google.com/file/d/1tFAi_s0MVsgtea0CVX1bM24AiP0G6Cf3/view?usp=sharing